europass



Md. Shifaul Hasan

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ABOUT ME

Computer Science and Engineering graduate (CGPA 3.52/4.00) with hands-on deep learning research in medical image analysis and speech emotion recognition. Co-authored three publications, including a first-author paper in Digital Health (SAGE) that proposes a Tri-Path Attention Stacked Ensemble, which achieved 94.44% accuracy on HAM10000 skin lesion classification with Grad-CAM interpretability, and a Springer conference paper on LSTM-based speech emotion recognition. Proficient in Python, PyTorch, TensorFlow and Keras, with additional full-stack experience in Laravel and MySQL. Seeking a Master's programme in Artificial Intelligence, Machine Learning or Data Science to continue research on reliable, interpretable AI for healthcare.

EDUCATION AND TRAINING

25/12/2021 – 04/04/2026 Dhaka, Bangladesh

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING IUBAT – International University of Business Agriculture and Technology

CGPA: 3.52 / 4.00 (A+ = 4.00); 144 credits; medium of instruction: English
- Relevant coursework: Applied Calculus (A+), Statistics (A-), Discrete Mathematics (A-), Linear Algebra, Numerical Analysis, Data Structures and Algorithms (A-), Theory of Computation (A-), Database Management Systems (A), Software Engineering (A+), Operating Systems, Computer Graphics (A), Compiler Design (A-), Data Communication and Computer Networks (A-), Programming in C, C++ and Java (OOP)
- Bachelor's Thesis (CSC 488, 6 credits, A+): Hierarchical Attention Stacked Ensemble with Matthews-Correlation-Coefficient Weighted Averaging: A Novel Framework for Skin Lesion Classification (Supervisor: Anwar Hossain Efat). The research led to two publications in Digital Health (SAGE).
- Practicum (CSC 490, 6 credits, A+): Mini E-Commerce – Inventory Management & Smart Billing System, built with Laravel and MySQL (Supervisor: Dr. Md. Alomgir Hossain)
- Research focus: deep learning for medical image classification and speech emotion recognition; three co-authored publications
- Technical skills: Python, PyTorch, TensorFlow, Keras, Laravel, Django, MySQL

01/01/2018 – 01/01/2020 Bogura, Bangladesh

HIGHER SECONDARY CERTIFICATE (HSC) Armed Police Battalion Public School & College, Bogura

Group: Science
Board: Rajshahi Education Board
Result: GPA 5.00 / 5.00 (5.00 even without the additional subject)
Subjects: Bangla, English, Information and Communication Technology (ICT), Physics, Chemistry, Biology
Additional (4th) subject: Higher Mathematics
Highlights: A+ (GP 5.00) in all subjects, including the additional subject Higher Mathematics.
Exam year: 2020

01/01/2016 – 31/12/2018 Dinajpur, Bangladesh

SECONDARY SCHOOL CERTIFICATE (SSC) Daudpur High School

Group: Science
Board: Dinajpur Education Board
Result: GPA 5.00 / 5.00

Subjects: Bangla, English, Mathematics, Bangladesh and Global Studies, Islam and Moral Education, Physics, Chemistry, Biology, Information and Communication Technology (ICT)
Additional (4th) subject: Higher Mathematics (A+)
Highlights: A+ in Mathematics, Higher Mathematics, Chemistry, Biology and ICT.
Exam year: 2018

WORK EXPERIENCE

WEB DEVELOPMENT INTERN – BANGLA SOFT COMPUTER – 01/10/2025 – 13/01/2026 – DHAKA, BANGLADESH

- Developed an inventory management and smart billing system using Laravel and MySQL, following the MVC architecture
- Implemented role-based authentication (Laravel Breeze) separating admin and customer users
- Built admin features for product and category CRUD and smart billing data, and customer features for catalog browsing and order placement using Eloquent ORM transactions
- Worked both independently and in a team; rated excellent in performance and conduct in the internship certificate

PRIVATE TUTOR – SELF-EMPLOYED (PRIVATE TUITION) – 10/01/2022 – 28/12/2024 – DHAKA, BANGLADESH

- Taught students privately, explaining lessons clearly and adapting explanations to each student's pace
- Supported student learning with patience and regular guidance
- Developed communication, planning and mentoring skills

SKILLS

Good communicator and team player | Able to work independently and in a team | Responsibility and time management

Programming:

Python | C | Java | C++ | PHP | JavaScript

Web Development:

HTML5 | Responsive Web Design | Laravel | Django | CSS3

Machine Learning & Deep Learning:

TensorFlow | Keras

ML Techniques:

Supervised Learning | Classification | Ensemble Learning

Computer Vision:

Feature Extraction & Selection | Medical Image Classification | Image Processing

Databases:

Database Design | MySQL

Software Engineering:

OOP | Data Structures & Algorithms | MVC Architecture

Development Tools:

Kaggle | Git | GitHub | Jupyter Notebook | Visual Studio Code | Google Colab

Document & Presentation Tools:

LaTeX | Microsoft Word | Microsoft PowerPoint | Microsoft Excel

LANGUAGE SKILLS

Mother tongue(s): **BANGLA**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PUBLICATIONS

2026

Enhancing skin lesion classification using a Tri-Path Attention Stacked Ensemble architecture with Cohen's Kappa Proportioned Averaging

Developed a Tri-Path Attention Stacked Ensemble (TASE) framework for 7-class skin lesion classification on the HAM10000 dataset (10,015 dermoscopic images). The model uses EfficientNet backbones with triplet attention (soft, channel and squeeze-excitation) and a novel Cohen's Kappa proportioned averaging method for multi-layer ensemble fusion. Achieved 94.44% accuracy, with Grad-CAM used for interpretability. First author: contributed to methodology, software implementation, validation, formal analysis, investigation and the original manuscript draft.

Hasan, M. S., Efat, A. H., Bhuiyan, J. A., & Mansia, F. M. (2026). Enhancing skin lesion classification using a Tri-Path Attention Stacked Ensemble architecture with Cohen's Kappa Proportioned Averaging. Digital Health, 12

Journal Name: Digital Health | **Volume, Issue and Pages**: Vol. 12 | **Publisher**: SAGE Publications

Link https://doi.org/10.1177/20552076261462026

2026

Hierarchical attention stacked ensemble with Matthews-correlation-coefficient weighted averaging: A novel framework for skin lesion classification

Proposed a Hierarchical Attention Stacked Ensemble (HASE) with MCC-weighted averaging for 7-class skin lesion classification on HAM10000. Built on EfficientNet backbones with triplet attention. Achieved 93.96% accuracy, with Grad-CAM interpretability. Based on my bachelor's thesis; co-author.

Bhuiyan, J. A., Efat, A. H., Hasan, M. S., & Mansia, F. M. (2026). Hierarchical attention stacked ensemble with Matthews-correlation-coefficient weighted averaging: A novel framework for skin lesion classification. Digital Health, 12.

Journal Name: Digital Health | **Volume, Issue and Pages**: Vol. 12 | **Publisher**: SAGE Publications

Link https://doi.org/10.1177/20552076261433750

Emotion Recognition from Speech Using LSTM-Based Deep Learning Model

Co-authored a speech emotion recognition paper using an LSTM-based deep learning model. Presented at BIM 2025, Dhaka International University (25-27 September 2025), and published by Springer.

Bhuiyan, J. A., Imran, F., Rony, I. A., Anni, F. A., & Hasan, M. S. (2026). Emotion Recognition from Speech Using LSTM-Based Deep Learning Model. In 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025). Springer.

Journal Name: 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025) | **Publisher**: Springer

Link https://doi.org/10.1007/978-3-032-15346-3_32

CERTIFICATIONS

ICT Division, Government of Bangladesh (EDGE Project), 22/07/2025

Python (Django) Course Completion

Completed a course on web application development with Python and the Django framework, under the EDGE Project of the ICT Division, Government of Bangladesh.

Mode of learning: Hybrid

Bangla Soft Computer, 13/01/2026

Web Development Internship Certificate

Completed a web development internship (1 October 2025 to 13 January 2026), developing an inventory management and smart billing system using Laravel and MySQL (MVC architecture). Rated excellent in performance and conduct, with good communication skills and the ability to work independently and in a team.

Mode of learning: Work based

Simplilearn SkillUp, 01/04/2025

Machine Learning Course Completion Certificate

Foundational training in Machine Learning algorithms and concepts, completed through Simplilearn SkillUp.

Mode of learning: Online

Kaggle, 15/04/2025

Python Coder (Badge Certificate)

Kaggle badge awarded for proficiency in core Python programming concepts.

Mode of learning: Online

Link https://www.kaggle.com/certification/badges/mdshifaulhasan/30

HP LIFE (HP Foundation), 20/03/2025

AI for Beginners

Introductory course on key AI concepts, the importance of data, and AI applications in business, offered by HP LIFE (HP Foundation).

Mode of learning: Online

Link https://www.life-global.org/certificate/2fb18610-49d1-444d-ac22-1fa7f68e7610

FutureNation (Grameenphone & UNDP initiative), 17/03/2025

Data Analysis with Excel

Course on data analysis, data manipulation and reporting using Excel, offered by FutureNation (Grameenphone & UNDP initiative).

Mode of learning: Online

Dhaka International University

3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025)

Awarded for the successful presentation of the paper “Emotion Recognition from Speech Using LSTM-Based Deep Learning Model” (Paper ID: 453) at the 3rd International Conference on Big Data, IoT and Machine Learning, held on 25-27 September 2025 at Dhaka International University, Bangladesh. The paper was co-authored and later published by Springer.

Mode of learning: Research-Lab based

Link https://link.springer.com/chapter/10.1007/978-3-032-15346-3_32